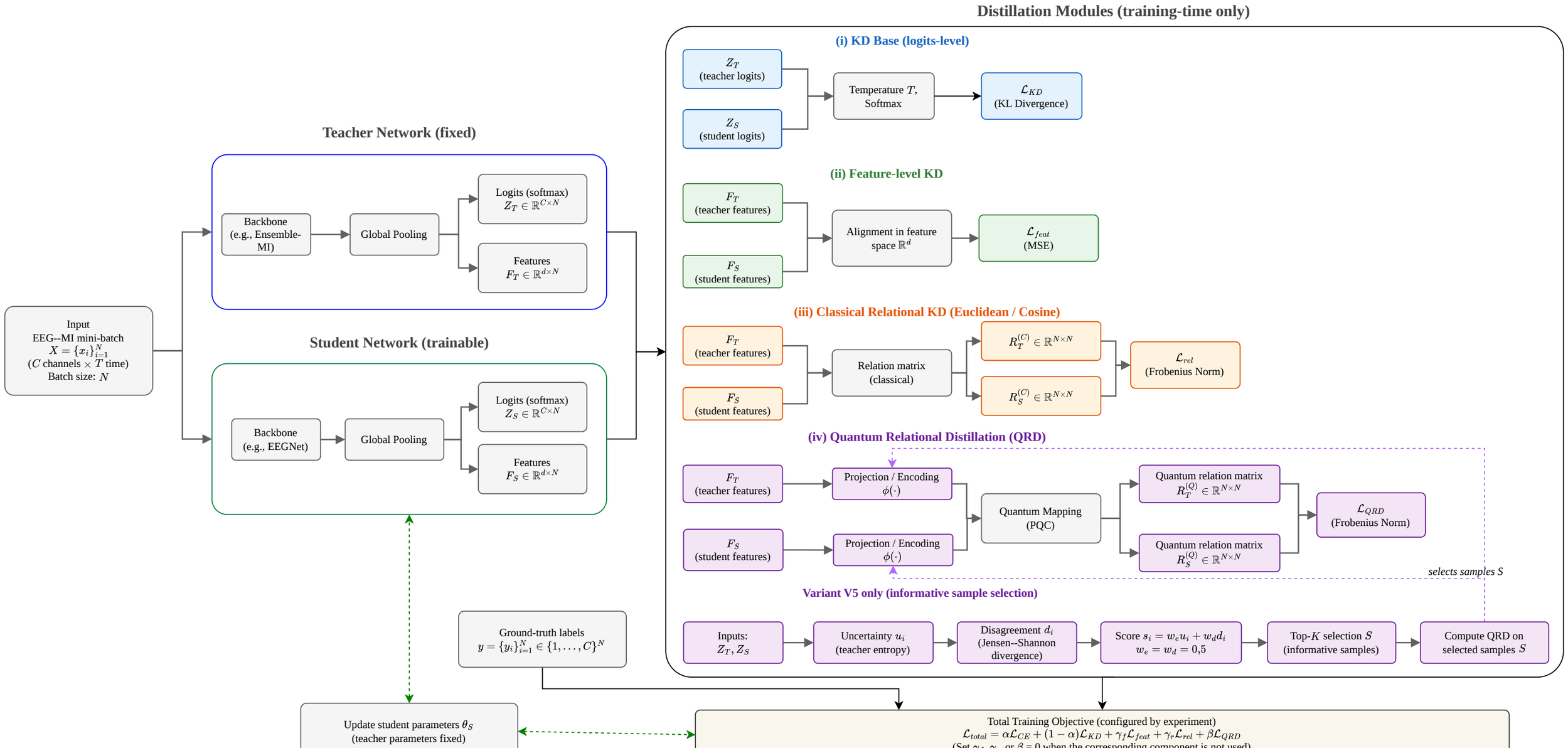

Distillation Modules (training-time only)
Teacher Network (fixed)
Backbone (e.g., Ensemble-MI)
Global Pooling
Logits (softmax) $Z_T \in \mathbb{R}^{C\times N}$
Features $F_T \in \mathbb{R}^{d\times N}$
Input EEG--MI mini-batch $X = \{x_i\}_{i=1}^N$ (C channels $\times$ T time) Batch size: N
Student Network (trainable)
Backbone (e.g., EEGNet)
Global Pooling
Logits (softmax) $Z_S \in \mathbb{R}^{C\times N}$
Features $F_S \in \mathbb{R}^{d\times N}$
(i) KD Base (logits-level)
Z_T (teacher logits)
Z_S (student logits)
Temperature T, Softmax
$\mathcal{L}_{KD}$ (KL Divergence)
(ii) Feature-level KD
F_T (teacher features)
F_S (student features)
Alignment in feature space $\mathbb{R}^d$
$\mathcal{L}_{feat}$ (MSE)
(iii) Classical Relational KD (Euclidean / Cosine)
F_T (teacher features)
F_S (student features)
Relation matrix (classical)
$R_T^{(C)} \in \mathbb{R}^{N\times N}$
$R_S^{(C)} \in \mathbb{R}^{N\times N}$
$\mathcal{L}_{rel}$ (Frobenius Norm)
(iv) Quantum Relational Distillation (QRD)
F_T (teacher features)
F_S (student features)
Projection / Encoding $\phi(\cdot)$
Projection / Encoding $\phi(\cdot)$
Quantum Mapping (PQC)
Quantum relation matrix $R_T^{(Q)} \in \mathbb{R}^{N\times N}$
Quantum relation matrix $R_S^{(Q)} \in \mathbb{R}^{N\times N}$
$\mathcal{L}_{QRD}$ (Frobenius Norm)
selects samples S
Variant V5 only (informative sample selection)
Inputs: Z_T, Z_S
Uncertainty u_i (teacher entropy)
Disagreement d_i (Jensen--Shannon divergence)
Score $s_i = w_e u_i + w_d d_i$ $w_e = w_d = 0{,}5$
Top-K selection S (informative samples)
Compute QRD on selected samples S
Ground-truth labels $y = \{y_i\}_{i=1}^N \in \{1, \ldots, C\}^N$
Update student parameters θ_S (teacher parameters fixed)
Total Training Objective (configured by experiment)
$\mathcal{L}_{total} = \alpha\mathcal{L}_{CE} + (1-\alpha)\mathcal{L}_{KD} + \gamma_f\mathcal{L}_{feat} + \gamma_r\mathcal{L}_{rel} + \beta\mathcal{L}_{QRD}$
(Set γ_f, γ_r, or β = 0 when the corresponding component is not used)